

Milestone 3 Report

Temporal Incident Analytics & Incident Response Performance

1. Objective

The objective of Milestone 3 was to extend the existing security analytics solution by analyzing **temporal incident patterns and incident response performance**. The dashboard was designed to identify when incidents occur most frequently, detect changes and activity spikes over time, analyze emerging threat patterns, and evaluate response performance across severity levels, locations, and operational coordinators.

2. Dashboard

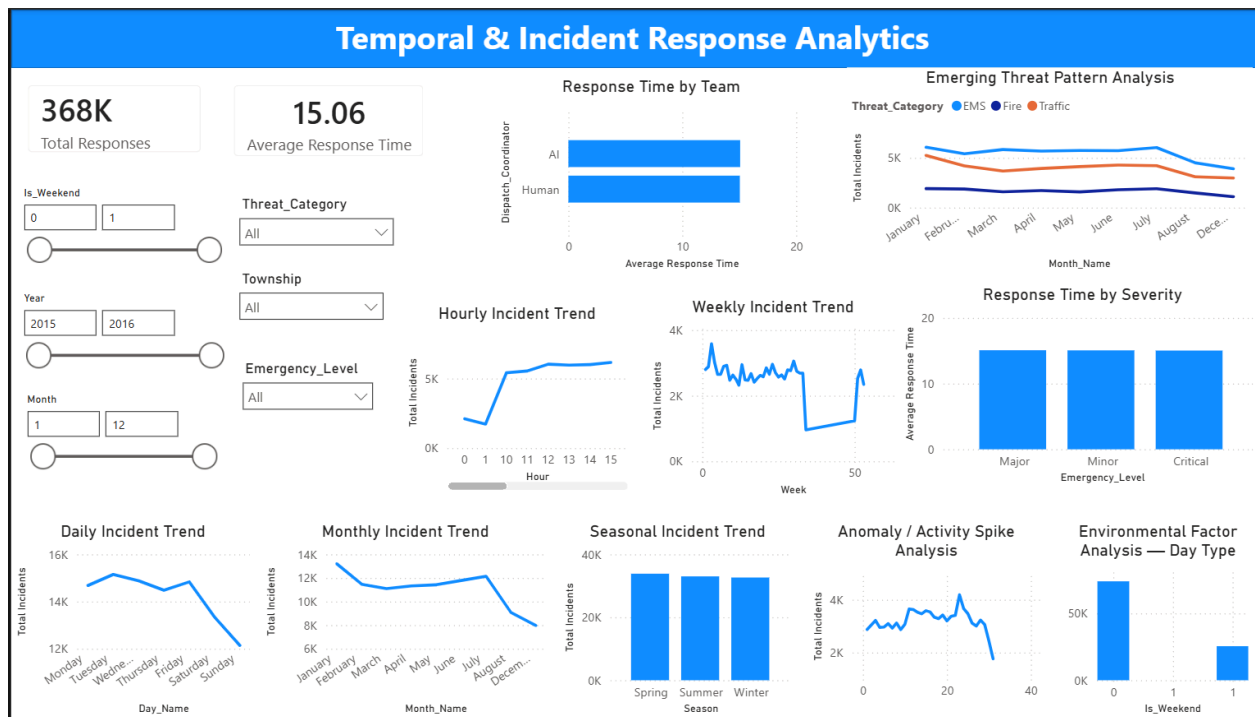
The Milestone 3 dashboard consists of two major analytical areas: **Temporal Incident Analytics** and **Incident Response Analytics**.

The temporal analysis includes **Hourly, Daily, Weekly, Monthly, and Seasonal Incident Trends**, along with **Activity Spike Analysis, Environmental Factor Analysis, and Emerging Threat Pattern Analysis**.

The response analysis includes **Total Responses, Average Response Time, Response Time by Dispatch Coordinator, Response Time by City, and Response Time by Severity**.

Interactive slicers for **Year, Month, Severity, Category, City, and Environmental Factor** allow stakeholders to dynamically explore the results.

Dashboard Screenshot:



Link: https://drive.google.com/file/d/1bgpZJXS4rRVdIM4t2enYhsV_zlo2N3vt/view?usp=drive_link

3. Key Insights

Insight 1 — Temporal Incident Patterns:

Hourly, daily, weekly, and monthly analysis provides a multi-level view of incident activity. This helps identify periods of increased operational demand and supports better resource planning.

Insight 2 — Activity Spikes and Emerging Patterns:

The activity-spike analysis highlights periods where incident volume rises significantly, while threat-category trends help identify categories whose activity changes over time.

Insight 3 — Response Performance:

Average response time and response-time comparisons across severity levels, cities, and dispatch coordinators provide visibility into operational response performance and help identify areas requiring improvement.

4. Business Recommendations

Recommendation 1 — Optimize Resource Scheduling:

Use identified peak hours, high-activity days, and high-demand periods to adjust staffing, emergency resources, and operational readiness.

Recommendation 2 — Improve Response Performance:

Locations or operational units with comparatively higher response times should be investigated for resource constraints, distance-related challenges, or other operational bottlenecks.

5. Data Limitations

The source data did not contain reliable **resolution timestamps or dedicated incident-status fields**, so Average Resolution Time, Open/Closed Incident analysis, and Response Status Distribution were not artificially generated. This ensures that the dashboard remains based on valid source information.